$$\frac{dx_1}{dt} = -x_1 + p_2(1+x_1)\, e^{\frac{x_2}{1+\frac{x_2}{20}}}$$

$$\frac{dx_2}{dt} = -x_2 + p_2 p_4 (1-x_1)\, e^{\frac{x_2}{1+\frac{x_2}{20}}} - p_5 x_2 =$$

$$= -(1+p_5)x_2 + p_2 p_4 (1-x_1)\, e^{\frac{x_2}{1+\frac{x_2}{20}}}$$

Пусть $E(x_2) = e^{\frac{x_2}{1+\frac{x_2}{20}}} = e^{\frac{20x_2}{20+x_2}}$

$$\frac{dx_1}{dt} = -x_1 + p_2(1-x_1)\, E(x_2)$$

$$\frac{dx_2}{dt} = -(1+p_5)x_2 + p_2 p_4 (1-x_1)\, E(x_2)$$

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

$$E'(x_2) = E(x_2)\cdot\left(\frac{20x_2}{20+x_2}\right)' = E(x_2)\,\frac{20(20+x_2)-20x_2}{(20+x_2)^2} =$$

$$= E(x_2)\,\frac{400}{(20+x_2)^2}$$

$$\frac{\partial f_1}{\partial x_1} = -1 - p_2 E(x_2) \qquad \frac{\partial f_1}{\partial x_2} = p_2(1-x_1)\,E(x_2)\,\frac{400}{(20+x_2)^2}$$

$$\frac{\partial f_2}{\partial x_2} = -p_2 p_4 E(x_2) \qquad \frac{\partial f_2}{\partial x_2} = -(1+p_5) + p_2 p_4 (1-x_1)\,E(x_2)\cdot \frac{400}{(20+x_2)^2}$$

$$J = \begin{bmatrix} -1 - p_2\, e^{\frac{20x_2}{20+x_2}} & p_2(1-x_1)\, e^{\frac{20x_2}{20+x_2}}\cdot\frac{400}{(20+x_2)^2} \\ -p_2 p_4\, e^{\frac{20x_2}{20+x_2}} & -(1+p_5) + p_2 p_4 (1-x_1)\, e^{\frac{20x_2}{20+x_2}}\,\frac{400}{(20+x_2)^2} \end{bmatrix}$$